Vanishing in Nuttall's normalized determinant expansion

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Abstract

We prove Conjecture 3 of Nuttall's *Determinantal approach to a proof of the Riemann hypothesis*. In fact, its coefficients vanish whenever $m + 2i < r(r-1)$, a slightly larger range than requested. The proof is purely algebraic: the coefficient of the normalization parameter is a symmetric differential operator of controlled order applied to a squared Vandermonde.

Task 1: the statement and its normalization

Fix an integer $r > 2$ and put $R = r(r-1)$. In [1, Section 7.2], the coefficients $C(i; m, j)$ arise from the permutation expansion of

$$D(n,r) = \det_{1\le a,b\le r}\left[\frac{b_{n+b-a}}{\Gamma(2(n+b-a)+1)}\right], \qquad y = \frac{1}{2n}, \quad n > r.$$

Here b_h are moments; their values play no role in the coefficient identity. The normalization is

$$\overline{D}(n,r) = y^R\Gamma(2n+1)^r \prod_{h=1}^{r-1}\big((2n+2h)(2n+2h-1)\big)^{r-h} D(n,r).$$

Nuttall asks for $C(i; m, j) = 0$ when

$$0 \le i \le R/2 - 1, \qquad 0 \le m \le R - 2i - 2, \qquad 1 \le j \le np(m).$$

We specify these coefficients without integrals. Symmetrization below is the average over all permutations of the r variables. For $\sigma \in S_r$ define

$$\nu_a = \sigma(a) - a, \qquad M_\sigma(t) = \prod_{a=1}^{r} t_a^{2(\nu_a + r - 1)}, \tag{1}$$

$$Z_\sigma(y) = \prod_{a=1}^{r} \prod_{s=2(\sigma(a)-a)+1}^{2(r-a)} (1 + sy). \tag{2}$$

An empty product is 1. All exponents in (1) are nonnegative, and every Z_σ is a polynomial of degree at most R. Set

$$P_r(y,t) = \sum_{\sigma\in S_r} \operatorname{sgn}(\sigma) Z_\sigma(y)\, \operatorname{Sym} M_\sigma(t). \tag{3}$$

To check (2), write $z = 2n$. Multiplying row a of the gamma-ratio matrix by $\Gamma(z + 2(r-a) + 1)/\Gamma(z+1)$ and its column b by $y^{2(r-b)}$ gives the entry

$$y^{2(r-b)} \frac{\Gamma(z + 2(r-a) + 1)}{\Gamma(z + 2(b-a) + 1)} = \prod_{s=2(b-a)+1}^{2(r-a)} (1 + sy).$$

The product of these row and column factors is precisely the stated normalization of the gamma-ratio determinant. Thus Z_σ is its unsigned permutation component.

Write $t_a = 1 + x_a$. The source's symmetrized monomials $\mu(m, j)$ form a basis for symmetric homogeneous polynomials of degree m. Its equations (7.17)–(7.22) therefore say exactly that

$$P_r(y, 1 + x) = \sum_{i,m,j} C(i; m, j) y^i \mu(m, j)(x). \tag{4}$$

Indeed, replacing the source variables by $\tau(1 + x_a)$ extracts the common factor τ^{2R} from every M_σ, as in (7.19). This also fixes all normalizations of C.

Theorem 1. *For every $r \geq 1$, the coefficient of y^i in $P_r(y, 1+x)$ has no monomial of total x-degree less than $R - 2i$. Consequently*

$$C(i; m, j) = 0 \quad \textit{whenever} \quad m + 2i < R.$$

In particular, all the coefficients required by Conjecture 3 vanish.

Polynomial dependence of the weights

For any integer v define a formal power series at $y = 0$ by

$$F_v(y) = \begin{cases} \displaystyle\prod_{s=1}^{2v} (1 + sy)^{-1}, & v \geq 0, \\ \displaystyle\prod_{s=2v+1}^{0} (1 + sy), & v < 0. \end{cases}$$

Also put

$$K_r(y) = \prod_{h=1}^{r-1} \big((1 + 2hy)(1 + (2h-1)y)\big)^{r-h}.$$

Since $\sum_a \nu_a = 0$, cancellation of the finite products, or the gamma-ratio calculation above, gives

$$Z_\sigma(y) = K_r(y) \prod_{a=1}^{r} F_{\nu_a}(y). \tag{5}$$

Lemma 1. *The coefficient $[y^k]F_v(y)$ is a polynomial in v of degree at most $2k$, valid for positive, zero, and negative integers v. Hence there is a symmetric polynomial $Q_i(v_1, \ldots, v_r)$, of total degree at most $2i$, such that*

$$[y^i] Z_\sigma(y) = Q_i(\nu_1, \ldots, \nu_r).$$

Proof. Let $S_\ell(u)$ be the polynomial of degree $\ell+1$ satisfying $S_\ell(0)=0$ and $S_\ell(u)-S_\ell(u-1)=u^\ell$. Such a polynomial exists because the finite-difference operator takes degree $\ell+1$ to degree ℓ with nonzero leading coefficient. Telescoping identifies it with $\sum_{s=1}^{u} s^\ell$ for positive integers u, and with $-\sum_{s=u+1}^{0} s^\ell$ for negative integers u. Thus, in both cases,

$$\log F_v(y)=\sum_{\ell\ge 1}\frac{(-1)^\ell}{\ell}S_\ell(2v)y^\ell.$$

In the coefficient of y^k in its exponential, each product has $\sum_\ell \ell a_\ell = k$ and degree at most $\sum_\ell(\ell+1)a_\ell \le 2k$ in v. Multiplication over the r variables preserves the total-degree bound and gives a symmetric polynomial. The scalar series $K_r(y)$ preserves both properties, proving the assertion. □

Symmetrization and differential order

Define commuting operators

$$E_a=\frac{1}{2}t_a\frac{\partial}{\partial t_a}-(r-1),\qquad 1\le a\le r.$$

Equation (1) gives $E_aM_\sigma=\nu_aM_\sigma$. Since Q_i is symmetric, $Q_i(E_1,\dots,E_r)$ commutes with symmetrization. Therefore

$$[y^i]P_r(y,t)=Q_i(E_1,\dots,E_r)B_r(t),\qquad B_r(t)=\mathrm{Sym}\sum_{\sigma\in S_r}\mathrm{sgn}(\sigma)M_\sigma(t). \tag{6}$$

The operator on the right has differential order at most $2i$.

Put $u_a=t_a^2$ and $V(u)=\prod_{a<b}(u_b-u_a)$. The determinant expansion shows that

$$\sum_{\sigma\in S_r}\mathrm{sgn}(\sigma)M_\sigma(t)=\left(\prod_{a=1}^{r}u_a^{r-a}\right)V(u).$$

When we symmetrize, $V(u)$ contributes the sign of the variable permutation. The resulting alternating sum of the descending powers $r-1,r-2,\dots,0$ is $(-1)^{R/2}V(u)$. Hence

$$B_r(t)=\frac{(-1)^{R/2}}{r!}\prod_{a<b}(t_b^2-t_a^2)^2. \tag{7}$$

Proof of Theorem 1. At $t=1+x$ every factor in (7) has the form

$$(t_b^2-t_a^2)^2=(x_b-x_a)^2(2+x_a+x_b)^2.$$

Thus $B_r(1+x)$ belongs to the Rth power of the ideal $\mathfrak{m}=(x_1,\dots,x_r)$. A differential operator of order at most d with polynomial coefficients maps $\mathfrak{m}^R$ into $\mathfrak{m}^{\max(R-d,0)}$: each derivative lowers total degree by at most one, and multiplication by a polynomial cannot lower it. Apply this with $d=2i$ to (6). All homogeneous components of degree $m<R-2i$ vanish. Their coefficients in the independent basis $\mu(m,j)$ in (4) therefore vanish individually. □

The proof is valid for arbitrary formal moments and uses no analytic assumption on Nuttall's function Φ. Its conclusion is the algebraic coefficient identity. The analytic positivity and asymptotic questions motivating the source require additional arguments.

Verification and proposed follow-up tasks

The accompanying standard-library Python verifier directly expands the permutation products (2) and the symmetrized binomial coefficients. It checks 2,810 zero coefficients in ranks 2–5, including the extra degree covered by Theorem 1. It separately compares the product normalization with exact factorial ratios. These finite checks supplement the all-rank proof.

Task 2 (proposed). Give a closed formula for the part of $P_r(y, 1+x)$ of weighted degree R, with weights 2 for y and 1 for every x_a.

Task 3 (proposed). Determine the exact order of vanishing at $x = 0$ of $[y^i]P_r(y, 1+x)$ for every $0 \le i \le R/2$, including when the bound $R - 2i$ is attained.

Task 4 (proposed). Replace the gamma arguments $2(n+b-a)+1$ by $a_0(n+b-a)+b_0$, with integer $a_0 > 0$, and determine the optimal coefficient-vanishing range under the corresponding moment normalization.

Task 5 (proposed). For Nuttall's particular Φ, combine the vanishing identity with a saddle-point expansion to determine the first nonzero asymptotic term of $\overline{D}(n, r)$, with a proved remainder bound for each fixed r.

References

[1] J. Nuttall, *Determinantal approach to a proof of the Riemann hypothesis*, arXiv:1111.1128v1 (2011), Sections 6.5 and 7, especially Conjecture 3 and equations (7.15)–(7.24). `https://arxiv.org/abs/1111.1128`.

Review status. Author-agent review; no independent peer review or proof-assistant verification. Dated scope and claim checks are documented in the accompanying audits.